

Functional Dependencies, Minimal Cover & Normalization Proof

Farmer Cooperative Supply Chain System (FCCSS)

SECTION 1: ALL FUNCTIONAL DEPENDENCIES

1. FARMER

FarmerID → AadhaarNo

FarmerID → Name

FarmerID → Phone

FarmerID → Village

FarmerID → Status

AadhaarNo → FarmerID (AadhaarNo is unique per farmer)

AadhaarNo → Name, Phone, Village, Status

2. COOPERATIVE_SOCIETY

CoopID → Name

CoopID → RegNo

CoopID → District

CoopID → State

CoopID → Phone

RegNo → CoopID (registration number is unique)

RegNo → Name, District, State, Phone

3. FARMER_MEMBERSHIP

MemberID → FarmerID

MemberID → CoopID

MemberID → EnrollDate

MemberID → MemberFee

MemberID → Status

(FarmerID, CoopID) → MemberID (a farmer joins a coop only once)

(FarmerID, CoopID) → EnrollDate, MemberFee, Status

4. CROP_LOT

LotID → FarmerID

LotID → CropName

LotID → Variety

LotID → QtyKg

LotID → HarvestDate

LotID → Season

5. PROCUREMENT

ProcureID → LotID

ProcureID → CoopID

ProcureID → ProcureDate

ProcureID → QtyKg

ProcureID → PricePerKg

ProcureID → TotalAmt (derived: QtyKg × PricePerKg, stored for audit)

6. QUALITY_CHECK

QCID → LotID

QCID → Grade

QCID → MoisturePct

QCID → Inspector

QCID → QCDate

7. FARMER_PAYMENT

PaymentID → ProcureID

PaymentID → FarmerID

PaymentID → GrossAmt

PaymentID → Deductions

PaymentID → NetPayable (derived: GrossAmt - Deductions, stored for audit)

PaymentID → PayDate

PaymentID → Mode

8. WAREHOUSE

WarehouseID → CoopID

WarehouseID → Location

WarehouseID → CapacityKg

WarehouseID → Address

9. INVENTORY

InventoryID → WarehouseID

InventoryID → ProcureID

InventoryID → CropName

InventoryID → Grade

InventoryID → QtyAvailKg

InventoryID → ReceivedDate

InventoryID → ShelfDate

10. LOGISTICS_PARTNER

PartnerID → Name

PartnerID → ServiceRegion

PartnerID → GSTIN

PartnerID → Phone

GSTIN → PartnerID (GSTIN is nationally unique)

GSTIN → Name, ServiceRegion, Phone

11. VEHICLE

VehicleID → PartnerID

VehicleID → VehicleType

VehicleID → CapacityKg

VehicleID → RegNo

RegNo → VehicleID (registration number is unique)

RegNo → PartnerID, VehicleType, CapacityKg

12. TRANSPORT_JOB

JobID → VehicleID

JobID → ProcureID

JobID → OrderID

JobID → SchedDate

JobID → SrcAddr

JobID → AmtPayable

JobID → Status

13. BUYER

BuyerID → BizName
BuyerID → Type
BuyerID → Email
BuyerID → GSTIN
GSTIN → BuyerID (GSTIN is nationally unique)
GSTIN → BizName, Type, Email

14. ORDER_HEADER

OrderID → BuyerID
OrderID → OrderDate
OrderID → Status
OrderID → DelivAddr
OrderID → QtyOrdered

15. ORDER_ITEM

OrderItemID → OrderID
OrderItemID → InventoryID
OrderItemID → QtyKg
OrderItemID → PricePerKg
OrderItemID → TotalAmt (derived: QtyKg × PricePerKg, stored for audit)
(OrderID, InventoryID) → OrderItemID (same item ordered once per order)
(OrderID, InventoryID) → QtyKg, PricePerKg, TotalAmt

16. CROP_PRICE

PriceID → CoopID
PriceID → CropName
PriceID → Grade
PriceID → PricePerKg
PriceID → EffectiveDate
PriceID → SettleAdmin
(CoopID, CropName, Grade, EffectiveDate) → PriceID (price set once per crop/grade/date)
(CoopID, CropName, Grade, EffectiveDate) → PricePerKg, SettleAdmin

17. SUBSIDY_SCHEME

SchemeID → SchemeName
SchemeID → Type
SchemeID → StartDate
SchemeID → EligibleCrop

18. SUBSIDY_DISBURSEMENT

DisburseID → SchemeID
DisburseID → FarmerID
DisburseID → ProcureID
DisburseID → AmtDisbursed
DisburseID → DisDate
DisburseID → RefNo
DisburseID → Status

SECTION 2: MINIMAL FD SET (CANONICAL COVER)

The canonical cover (minimal FD set) is obtained by: (a) decomposing all RHS into single attributes, (b) removing extraneous attributes from LHS, and (c) eliminating redundant FDs. Derived/computed attributes

(TotalAmt, NetPayable) are excluded because they are not independent FDs — they follow from arithmetic, not relational dependency.

1. FARMER — Canonical Cover

FarmerID $\rightarrow$ AadhaarNo

FarmerID $\rightarrow$ Name

FarmerID $\rightarrow$ Phone

FarmerID $\rightarrow$ Village

FarmerID $\rightarrow$ Status

AadhaarNo $\rightarrow$ FarmerID

Note: AadhaarNo $\rightarrow$ {Name, Phone, Village, Status} are derivable via AadhaarNo $\rightarrow$ FarmerID $\rightarrow$ {...}, so they are redundant and removed.

2. COOPERATIVE_SOCIETY — Canonical Cover

CoopID $\rightarrow$ Name

CoopID $\rightarrow$ RegNo

CoopID $\rightarrow$ District

CoopID $\rightarrow$ State

CoopID $\rightarrow$ Phone

RegNo $\rightarrow$ CoopID

Note: RegNo $\rightarrow$ {Name, District, State, Phone} are derivable via RegNo $\rightarrow$ CoopID $\rightarrow$ {...}, so they are redundant and removed.

3. FARMER_MEMBERSHIP — Canonical Cover

MemberID $\rightarrow$ FarmerID

MemberID $\rightarrow$ CoopID

MemberID $\rightarrow$ EnrollDate

MemberID $\rightarrow$ MemberFee

MemberID $\rightarrow$ Status

(FarmerID, CoopID) $\rightarrow$ MemberID

Note: (FarmerID, CoopID) $\rightarrow$ {EnrollDate, MemberFee, Status} removed; derivable via (FarmerID, CoopID) $\rightarrow$ MemberID $\rightarrow$ {...}.

4. CROP_LOT — Canonical Cover

LotID $\rightarrow$ FarmerID

LotID $\rightarrow$ CropName

LotID $\rightarrow$ Variety

LotID $\rightarrow$ QtyKg

LotID $\rightarrow$ HarvestDate

LotID $\rightarrow$ Season

5. PROCUREMENT — Canonical Cover

ProcureID $\rightarrow$ LotID

ProcureID $\rightarrow$ CoopID

ProcureID $\rightarrow$ ProcureDate

ProcureID $\rightarrow$ QtyKg

ProcureID $\rightarrow$ PricePerKg

Note: TotalAmt excluded (computed attribute, not a relational FD).

6. QUALITY_CHECK — Canonical Cover

QCID $\rightarrow$ LotID

QCID $\rightarrow$ Grade

QCID $\rightarrow$ MoisturePct

QCID → Inspector

QCID → QCDate

7. FARMER_PAYMENT — Canonical Cover

PaymentID → ProcureID

PaymentID → FarmerID

PaymentID → GrossAmt

PaymentID → Deductions

PaymentID → PayDate

PaymentID → Mode

Note: NetPayable excluded (computed attribute: GrossAmt - Deductions).

8. WAREHOUSE — Canonical Cover

WarehouseID → CoopID

WarehouseID → Location

WarehouseID → CapacityKg

WarehouseID → Address

9. INVENTORY — Canonical Cover

InventoryID → WarehouseID

InventoryID → ProcureID

InventoryID → CropName

InventoryID → Grade

InventoryID → QtyAvailKg

InventoryID → ReceivedDate

InventoryID → ShelfDate

10. LOGISTICS_PARTNER — Canonical Cover

PartnerID → Name

PartnerID → ServiceRegion

PartnerID → GSTIN

PartnerID → Phone

GSTIN → PartnerID

Note: GSTIN→{Name,ServiceRegion,Phone} removed; derivable via GSTIN→PartnerID→{...}.

11. VEHICLE — Canonical Cover

VehicleID → PartnerID

VehicleID → VehicleType

VehicleID → CapacityKg

VehicleID → RegNo

RegNo → VehicleID

Note: RegNo→{PartnerID,VehicleType,CapacityKg} removed; derivable via RegNo→VehicleID→{...}.

12. TRANSPORT_JOB — Canonical Cover

JobID → VehicleID

JobID → ProcureID

JobID → OrderID

JobID → SchedDate

JobID → SrcAddr

JobID → AmtPayable

JobID → Status

13. BUYER — Canonical Cover

BuyerID $\rightarrow$ BizName

BuyerID $\rightarrow$ Type

BuyerID $\rightarrow$ Email

BuyerID $\rightarrow$ GSTIN

GSTIN $\rightarrow$ BuyerID

Note: GSTIN $\rightarrow$ {BizName, Type, Email} removed; derivable via GSTIN $\rightarrow$ BuyerID $\rightarrow$ {...}.

14. ORDER_HEADER — Canonical Cover

OrderID $\rightarrow$ BuyerID

OrderID $\rightarrow$ OrderDate

OrderID $\rightarrow$ Status

OrderID $\rightarrow$ DelivAddr

OrderID $\rightarrow$ QtyOrdered

15. ORDER_ITEM — Canonical Cover

OrderItemID $\rightarrow$ OrderID

OrderItemID $\rightarrow$ InventoryID

OrderItemID $\rightarrow$ QtyKg

OrderItemID $\rightarrow$ PricePerKg

(OrderID, InventoryID) $\rightarrow$ OrderItemID

Note: TotalAmt excluded (computed); (OrderID, InventoryID) $\rightarrow$ {QtyKg, PricePerKg} removed as derivable via the above.

16. CROP_PRICE — Canonical Cover

PriceID $\rightarrow$ CoopID

PriceID $\rightarrow$ CropName

PriceID $\rightarrow$ Grade

PriceID $\rightarrow$ PricePerKg

PriceID $\rightarrow$ EffectiveDate

PriceID $\rightarrow$ SettleAdmin

(CoopID, CropName, Grade, EffectiveDate) $\rightarrow$ PriceID

Note: (CoopID, CropName, Grade, EffectiveDate) $\rightarrow$ {PricePerKg, SettleAdmin} removed; derivable via the composite key $\rightarrow$ PriceID $\rightarrow$ {...}.

17. SUBSIDY_SCHEME — Canonical Cover

SchemeID $\rightarrow$ SchemeName

SchemeID $\rightarrow$ Type

SchemeID $\rightarrow$ StartDate

SchemeID $\rightarrow$ EligibleCrop

18. SUBSIDY_DISBURSEMENT — Canonical Cover

DisburseID $\rightarrow$ SchemeID

DisburseID $\rightarrow$ FarmerID

DisburseID $\rightarrow$ ProcureID

DisburseID $\rightarrow$ AmtDisbursed

DisburseID $\rightarrow$ DisDate

DisburseID $\rightarrow$ RefNo

DisburseID $\rightarrow$ Status

SECTION 3: NORMALIZATION PROOF (BCNF)

A relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. Below, each relation's candidate keys and all determinants are listed to confirm this condition holds.

Relation	Candidate Key(s)	All Determinants in Minimal Cover	BCNF Argument
FARMER	{FarmerID}, {AadhaarNo}	FarmerID, AadhaarNo	Both determinants are candidate keys, hence superkeys. Every non-trivial FD has a superkey on LHS. BCNF ✓
COOPERATIVE_SOCIETY	{CoopID}, {RegNo}	CoopID, RegNo	Both are candidate keys. Every determinant is a superkey. BCNF ✓
FARMER_MEMBER_SHIP	{MemberID}, {FarmerID, CoopID}	MemberID, (FarmerID, CoopID)	Both determinants are candidate keys of the relation. Every non-trivial FD has a superkey on LHS. BCNF ✓
CROP_LOT	{LotID}	LotID	Single candidate key; LotID is the only determinant. BCNF ✓
PROCUREMENT	{ProcureID}	ProcureID	Single candidate key. Only determinant is a superkey. BCNF ✓
QUALITY_CHECK	{QCID}	QCID	Single candidate key. Only determinant is a superkey. BCNF ✓
FARMER_PAYMENT	{PaymentID}	PaymentID	Single candidate key. Only determinant is a superkey. BCNF ✓
WAREHOUSE	{WarehouseID}	WarehouseID	Single candidate key. Only determinant is a superkey. BCNF ✓
INVENTORY	{InventoryID}	InventoryID	Single candidate key. Only determinant is a superkey. BCNF ✓
LOGISTICS_PARTNER	{PartnerID}, {GSTIN}	PartnerID, GSTIN	Both are candidate keys. Every determinant is a superkey. BCNF ✓
VEHICLE	{VehicleID}, {RegNo}	VehicleID, RegNo	Both are candidate keys. Every determinant is a superkey. BCNF ✓
TRANSPORT_JOB	{JobID}	JobID	Single candidate key. Only determinant is a superkey. BCNF ✓
BUYER	{BuyerID}, {GSTIN}	BuyerID, GSTIN	Both are candidate keys. Every determinant is a superkey. BCNF ✓
ORDER_HEADER	{OrderID}	OrderID	Single candidate key. Only determinant is a superkey. BCNF ✓
ORDER_ITEM	{OrderItemID}, {OrderID, InventoryID}	OrderItemID, (OrderID, InventoryID)	Both determinants are candidate keys. Every non-trivial FD has a superkey on LHS. BCNF ✓
CROP_PRICE	{PriceID}, {CoopID, CropName, Grade, EffectiveDate}	PriceID, (CoopID, CropName, Grade, EffectiveDate)	Both determinants are candidate keys. Every non-trivial FD has a superkey on LHS. BCNF ✓
SUBSIDY_SCHEME	{SchemeID}	SchemeID	Single candidate key. Only determinant is a superkey. BCNF ✓
SUBSIDY_DISBURSEMENT	{DisburseID}	DisburseID	Single candidate key. Only determinant is a superkey. BCNF ✓

Conclusion

All 18 relations in the FCCSS database are in Boyce-Codd Normal Form (BCNF). In every relation, the only non-trivial determinants present in the minimal FD set are candidate keys of that relation. No partial dependencies or transitive dependencies on non-key attributes exist. Therefore the schema satisfies 1NF, 2NF, 3NF, and BCNF simultaneously.